

MOKOHINAU, New Zealand

WMO: 930690

Lat: 35.903S

Lon: 175.115E

Elev: 60

StdP: 100.61

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.5	10.3	2.2	4.5	10.9	3.6	4.9	11.5	20.4	13.1	19.0	12.9	8.4	180	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	3.6	23.5	19.8	22.7	19.3	22.2	19.0	21.4	22.3	20.6	21.7	20.0	21.3	4.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	15.8	21.8	20.2	15.0	21.3	19.6	14.4	20.9	62.4	22.4	59.8	21.8	57.7	21.3	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.3	16.7	14.7	DB	8.2	24.8	1.0	1.0	7.5	25.5	6.9	26.1	6.3	26.6	5.5	27.3	(4)
(5)			WB	5.2	21.8	0.9	1.0	4.6	22.5	4.1	23.1	3.6	23.6	3.0	24.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.4	19.7	20.4	19.6	18.0	16.1	14.1	13.3	13.1	13.8	14.7	16.1	18.2	(6)	
	DBStd	2.87	1.33	1.22	1.27	1.42	1.57	1.52	1.35	1.29	1.28	1.24	1.31	1.36	(7)	
	HDD10.0	1	0	0	0	0	0	0	1	1	0	0	0	0	(8)	
	HDD18.3	879	3	1	4	22	70	126	156	161	136	113	68	19	(9)	
	CDD10.0	2337	300	291	297	240	189	124	102	98	114	146	183	253	(10)	
	CDD18.3	173	45	58	43	12	1	0	0	0	0	0	1	14	(11)	
	CDH23.3	24	7	15	2	0	0	0	0	0	0	0	0	1	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	7.3	6.6	6.6	6.8	6.9	7.5	7.9	8.4	7.7	7.9	7.5	7.5	6.7	(14)	
(15) Precipitation	PrecAvg	1322	77	84	101	113	117	140	153	142	113	96	88	99	(15)	
	PrecMax	1797	229	256	215	226	274	250	394	274	296	175	218	236	(16)	
	PrecMin	863	22	23	35	38	34	57	32	56	57	31	39	26	(17)	
	PrecStd	240	48	54	48	48	54	50	74	42	45	35	39	52	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.2	24.7	23.5	21.8	20.0	18.1	17.0	16.5	17.4	18.4	20.7	22.7	(19)	
		MCWB	20.0	20.7	19.5	19.4	18.0	16.8	15.5	14.8	14.9	15.2	16.8	18.8	(20)	
	2%	DB	23.2	23.7	22.6	21.1	19.2	17.4	16.2	15.8	16.6	17.7	19.6	21.6	(21)	
		MCWB	19.5	20.1	19.1	18.5	17.3	15.8	14.4	14.1	14.5	14.9	16.3	18.1	(22)	
	5%	DB	22.5	23.0	22.0	20.5	18.7	16.8	15.7	15.4	16.1	17.1	18.8	20.9	(23)	
		MCWB	19.1	19.6	18.8	18.1	16.8	15.0	13.8	13.6	14.2	14.6	15.9	17.6	(24)	
	10%	DB	21.8	22.3	21.5	19.9	18.3	16.3	15.2	15.0	15.6	16.6	18.2	20.2	(25)	
		MCWB	18.8	19.3	18.5	17.7	16.3	14.3	13.3	13.1	13.8	14.3	15.6	17.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.9	22.5	21.4	20.9	19.0	17.4	16.2	15.7	16.1	16.6	18.2	20.4	(27)	
		MCDB	22.7	23.1	22.0	21.2	19.4	17.8	16.8	15.9	16.6	17.2	19.1	21.2	(28)	
	2%	WB	20.9	21.7	20.6	19.6	18.2	16.4	15.1	15.0	15.4	16.0	17.5	19.5	(29)	
		MCDB	22.1	22.5	21.6	20.3	18.8	17.1	15.8	15.5	16.1	16.8	18.4	20.5	(30)	
	5%	WB	20.2	21.0	19.9	19.0	17.5	15.7	14.5	14.3	15.0	15.6	17.0	18.9	(31)	
		MCDB	21.4	22.1	21.2	20.0	18.5	16.6	15.3	15.0	15.7	16.5	18.0	19.9	(32)	
	10%	WB	19.6	20.2	19.3	18.2	16.8	14.9	13.8	13.6	14.4	15.1	16.4	18.3	(33)	
		MCDB	21.0	21.6	20.8	19.5	18.1	16.0	15.0	14.6	15.4	16.3	17.7	19.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.8	3.6	3.2	2.8	2.4	2.3	2.3	2.5	2.9	3.3	3.7	3.7	(35)	
		MCDBR	4.4	4.4	3.7	3.1	2.5	2.5	2.5	2.7	3.2	3.6	4.3	4.5	(36)	
	5% WB	MCWBR	2.6	2.5	2.5	2.3	2.3	2.6	2.5	2.4	2.6	2.7	2.8	2.8	(37)	
		MCWBR	3.4	3.3	3.1	2.6	2.2	2.3	2.4	2.3	2.7	2.9	3.4	3.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.345	0.343	0.331	0.319	0.310	0.306	0.304	0.309	0.330	0.334	0.335	0.341	(40)	
		taud	2.597	2.632	2.652	2.677	2.667	2.655	2.656	2.647	2.562	2.563	2.586	2.616	(41)	
	Ebn at Noon		991	971	945	902	853	828	850	899	930	968	994	998	(42)	
		Edh at Noon	103	96	87	75	67	63	66	76	93	101	103	102	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.89	5.94	4.93	3.58	2.55	2.07	2.26	3.10	4.24	5.51	6.41	6.61	(44)	
	RadStd		0.39	0.41	0.32	0.24	0.16	0.14	0.13	0.18	0.21	0.33	0.38	0.55	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page